Goldman Sachs Research

Economics Research
27 October 2026 | 10:41AM HKT

ASIA ECONOMICS ANALYST

China's High-Tech Push: Support for Output, Less for Employment



Chelsea Song
+852-2978-0106 |
chelsea.song@gs.com
Goldman Sachs (Asia) L.L.C.

Xinquan Chen
+852-2978-2418 |
xinquan.chen@gs.com
Goldman Sachs (Asia) L.L.C.

- Industrial upgrading and technological self-reliance have long been central to China's development strategy. Amid intensifying technology competition with the US, policy has increasingly focused on high-tech and AI-related industries. In this report, we assess whether high-tech sectors can offset the property-sector drag, how far their growth supports employment and household income, and how much data center construction can boost GDP growth in the coming years.
- We define high-tech sectors as high-tech manufacturing (including pharmaceuticals and equipment manufacturing), new energy vehicles, and telecom, IT and software services. These sectors accounted for 38% of industrial production in 2025, up from 33% in 2021, while their share of total fixed asset investment rose from 13% to 18%. In contrast, their employment share increased only marginally, from 10.4% in 2018 to 10.8% in 2023 (latest data available), reflecting the capital-intensive nature of high-tech production.
- High-tech sectors tend to generate less domestic value added, labor income, and employment per yuan of final spending than either property or the broader services sector. Every RMB100 of final spending on high-tech goods and related services generates about RMB73 of domestic value added, compared with RMB90 for property and RMB92 for services sector. Within that, it generates around RMB33 of labor compensation, versus RMB40 for property and RMB46 for services. Job creation is similarly lower in high-tech sectors: each RMB1mn of final spending supports around 3 jobs, versus 4.6 in property and 4.4 across the services sector.
- We estimate that high-tech sectors contributed roughly 1.9pp to annual real GDP growth in 2016–25 and expect a broadly stable contribution of around 2pp annually in 2026–30. By our estimates, high-tech sectors, including upstream spillovers, account for 23% of GDP in 2026, higher than the property sector's 15% share. Although the property downturn is likely to persist, its drag should fade as the sector's GDP share declines. The combined growth impulse from high-tech growth and property contraction should become increasingly positive on net in 2027-30.

Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html.

- Data center build-out is a key component of China's AI push and an important part of the "Six Networks" investment initiative (water, power grids, computing, communications, urban underground pipelines, and logistics) outlined in the 15th Five-Year Plan (FYP). We estimate RMB28.4tn "Six Networks" total investment during the 15th FYP, 14% higher than during the 14th FYP. Within computing networks, we estimate data center build-out could add around 40bp to the average *level* of China's real GDP between 2026 and 2030.

China's High-Tech Push: Support for Output, Less for Employment

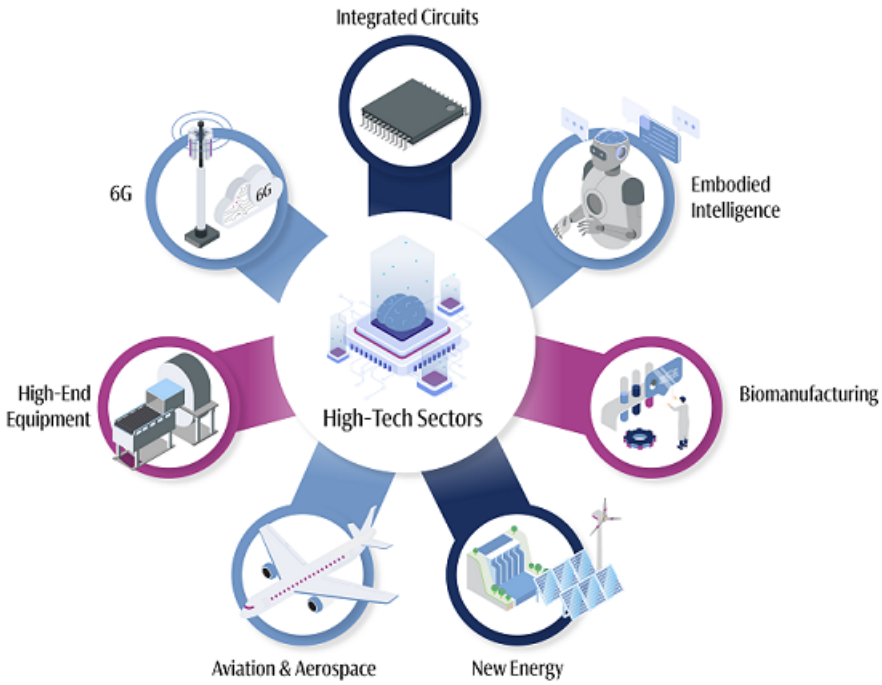
China's policymakers have long prioritized industrial upgrading and technological self-reliance as central to the country's development strategy. From the launch of "Made in China 2025" in 2015 to the emphasis on high-tech industries and AI in the 15th Five-Year Plan (FYP) for 2026–30, policy support has focused on strengthening domestic capabilities and global competitiveness in strategic industries, including semiconductors, advanced robotics, biotechnology, AI and related digital technologies (Exhibit 1).

This agenda has become more urgent amid intensifying technological competition with the US. Recent policy signals on advancing the integration of technological and industrial innovation underscores policymakers' intention to accelerate the commercial application of innovation and AI adoption. Meanwhile, the government has placed renewed emphasis on the broader "Six Networks" initiative – covering water networks, new-type power grids, computing-power networks, next-generation communications networks, urban underground pipeline networks, and logistics networks – as a means of supporting near-term investment while addressing longer-term infrastructure, resilience, and innovation needs.

For investors, the key questions are whether high-tech sectors can generate sufficient growth to offset the continuing drag from the property downturn, how far this expansion translates into employment and household income, and how large a near-term boost the "Six Networks" initiative, especially data center construction, could provide to investment and growth.

In this report, we build on our previous analysis of high-tech manufacturing by broadening the scope to include new energy vehicles (NEVs) manufacturing and telecom, IT and software services sectors. This expanded scope captures key parts of the AI supply chain, from semiconductors and computing equipment production to cloud computing and software services. Using an updated input-output framework, we estimate these sectors' contribution to real GDP growth and assess their implications for the labor market. We also size investment across the "Six Networks" during the 15th FYPs and gauge the growth impact from China's data center construction.

Exhibit 1: Policymakers' Unwavering Support for AI and Technology Sectors



Source: Goldman Sachs Global Investment Research

Stylized Facts on China's High-tech Sectors

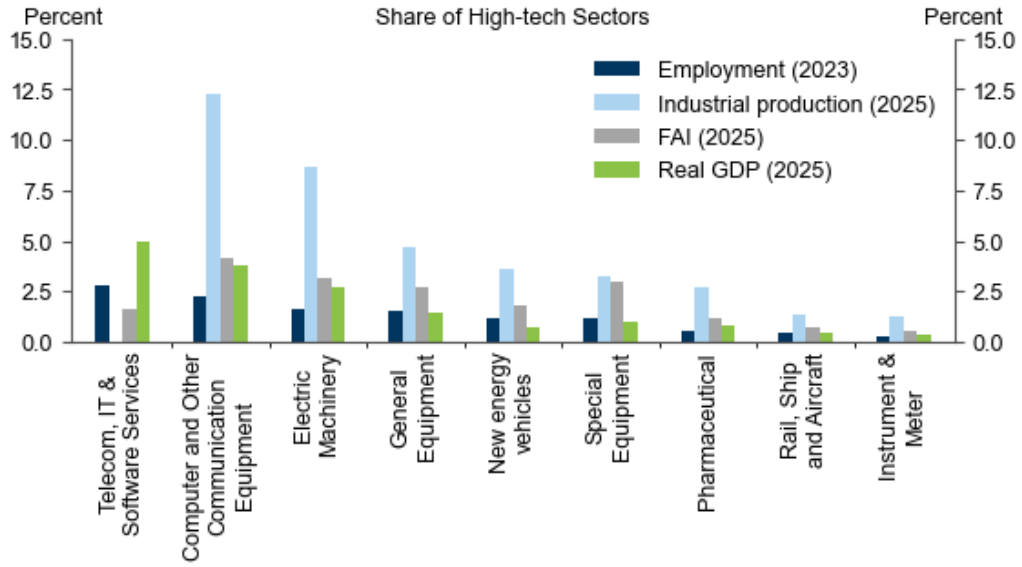
China's high-tech sectors have expanded rapidly over the past decade. Our expanded definition covers high-tech manufacturing—including computers, communications and electronics; electrical machinery, general and special equipment, pharmaceuticals and medical products, instruments and meters, and rail, ship and aircraft manufacturing—as well as new energy vehicle manufacturing and telecom, IT and software services. This scope broadly aligns with the NBS definition of high-tech sectors.

The manufacturing industries in this group accounted for around 38% of industrial production in 2025, up from 33% in 2021 (Exhibit 2). Over this period, their real output grew by an average of 8.3% per year, compared with 5.7% for headline industrial production. Investment in our expanded high-tech sectors, including both manufacturing and services, also grew faster than investment in the broader economy. FAI in these sectors increased by an average of 8.4% per year, compared with 4.7% for total FAI, raising their share of total investment from 13% in 2021 to 18.4% in 2025. Together, these high-tech sectors directly accounted for 16.4% of GDP in value-added terms in 2025, around 6pp more than a decade earlier.

However, these sectors have a much smaller employment footprint, accounting for only 10.8% of total employment in 2023, the latest year available, compared with 10.4% in 2018. Their composition also varies substantially across industries. Computer and communications equipment manufacturing accounts for the largest share of production and investment within the grouping but only 2.5% of employment, while telecom, IT and software services account for the largest share of employment (2.7%). These differences suggest that a larger contribution to production does not necessarily translate into a proportionate increase in employment. The implications for household income also

depend on how the gains from high-tech growth are distributed. We examine these links below.

Exhibit 2: High-Tech Sectors Account for 38% of Industrial Production and 18% of Investment, But Only 11% of Employment



Source: Haver Analytics, CEIC, Goldman Sachs Global Investment Research

High-tech Sector's Transmission to Growth and the Labor Market

Relative to the property sector, high-tech sectors' growing economic importance brings greater exposure to foreign demand and does not by itself necessitate a shift toward consumption-led growth.

China's latest input-output table shows that high-tech sectors rely primarily on domestic investment and external demand[1]. Only around 20% of spending on high-tech products and services comes from consumption by households and government, compared with roughly 70% for the overall services sector (Exhibit 3, left). Instead, nearly half of demand comes from purchases for investment, such as equipment and software, while foreign demand (i.e., exports) accounts for the remaining third. In this respect, the relatively small consumption share resembles that of the property sector, where demand is also dominated by investment, including households' purchases of new properties.

The main difference is high-tech's much greater exposure to foreign demand, particularly within high-tech manufacturing industries, where exports account for nearly half of final demand. A shift in activity from property toward high-tech therefore gives exports a greater role in China's growth outlook, even though high-tech sectors' own export dependence declined somewhat[2].

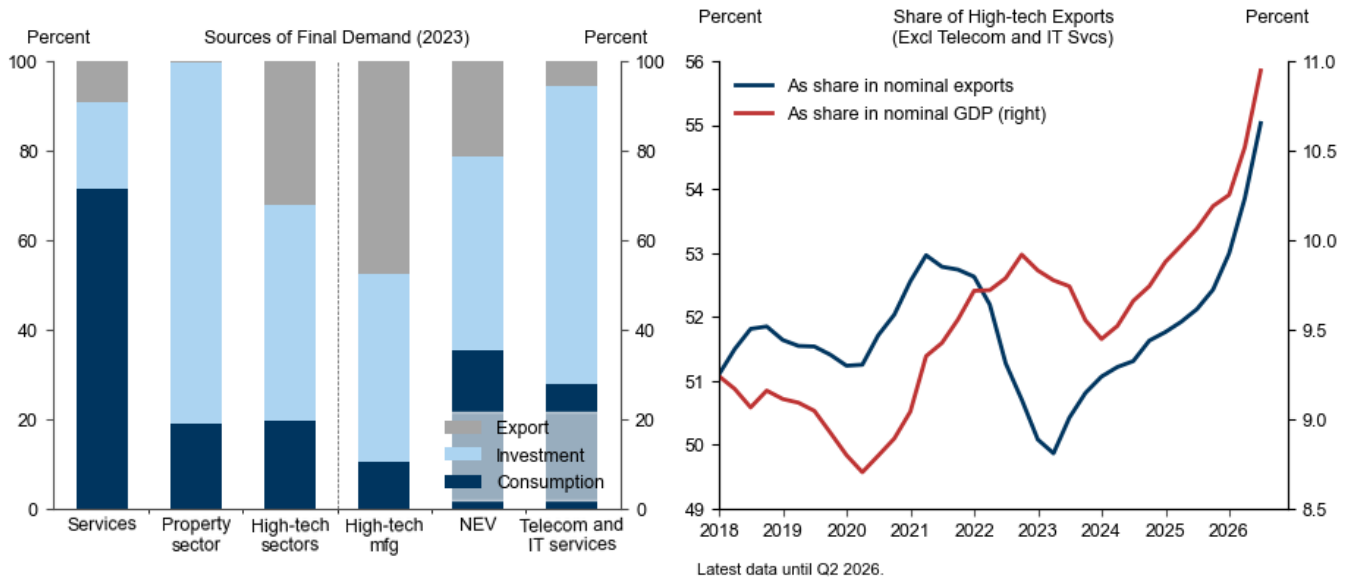
[1] These shares reflect the structure of expenditure in the 2023 input-output table (latest available). They cover purchases for consumption, investment and export, excluding goods and services used as intermediate inputs by domestic producers.

[2] Between 2017 and 2023, their export share of final spending fell by 3.8pp, while the consumption share rose by the same amount. However, this was driven by China's exit from its "zero Covid" policy in 2023 which led to a strong rebound in domestic demand.

High-tech manufacturing has also become a more important contributor to China's export performance. These industries' share within nominal goods exports increased by around 4pp between 2018 and 2025, while their gross exports rose by approximately 2pp of GDP over the same period (Exhibit 3, right). The shares have risen further in 2026, although that is mainly driven by higher prices from AI-related inflation in those high-tech sectors than in the rest of the economy.

Taken together, exports will be an important factor in sustaining high-tech production growth, alongside continued domestic investment. This reliance may partly explain policymakers' lackluster policy support for boosting household consumption.

Exhibit 3: High-tech Sector's Final Demand Is Investment- and Export-Led, with Greater Export Exposure than the Property Sector



Source: Goldman Sachs Global Investment Research, NBS, Trademap, Haver Analytics

Beyond their reliance on domestic investment and exports, high-tech sectors generate less domestic value added and support fewer jobs per yuan of final spending than either the property sector or the overall service sector. We estimate that every RMB100 of final spending on high-tech products and related services generates around RMB73 of domestic value added, including the value created by these industries and their domestic suppliers throughout the supply chain, with the rest leaking abroad through imported inputs. This compares with roughly RMB90 for the property sector – including construction and real estate services - and RMB92 for the overall services sector (Exhibit 4, left). The shortfall reflects high-tech sectors' greater reliance on imported inputs, meaning a larger portion of spending leaks into production abroad. Reducing import reliance is likely one of the reasons behind policymakers' intensified efforts to invest in home-grown high-tech development. Going forward, rising self-sufficiency in areas such as semiconductors will lift the domestic value-added share over time.

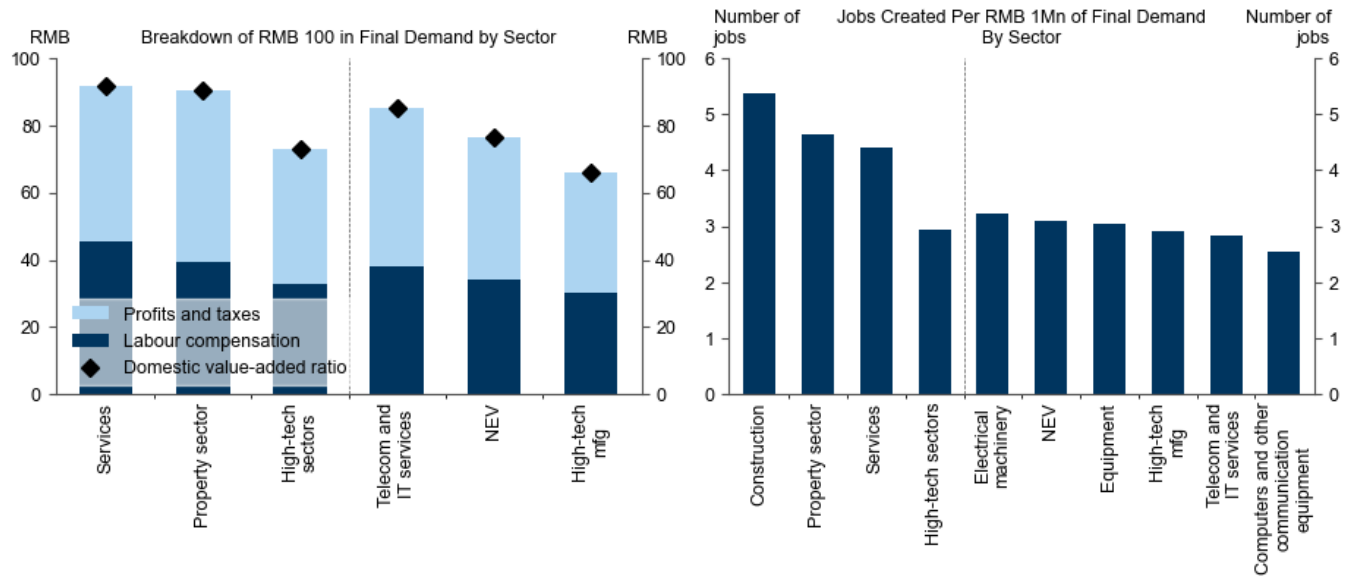
The smaller domestic value-added share also limits how much spending translates into labor income. Each RMB100 of final spending on high-tech output generates around RMB33 of labor compensation across these industries and their domestic suppliers, compared with RMB40 for property and RMB46 for the services sector. Relative to property, the gap mainly stems from greater import leakage, as workers in both sectors receive a similar share of domestic value added at around 45%. By contrast, the services

sector directs roughly 50% of domestic value added to workers, highlighting a much stronger pass-through from final spending to labor income.

Spending on high-tech sectors also supports fewer jobs, suggesting a limited boost to household consumption through employment. Each RMB1mn of final spending on high-tech sectors supports around 3 jobs across these industries and their domestic suppliers, compared with around 4.6 for property and 4.4 for services overall (Exhibit 4, right). This pattern extends beyond high-tech manufacturing sector, as telecom, IT and software services sector also support fewer jobs per yuan of spending than the broader services sector on average.

Taken together, these estimates suggest that an equivalent increase in final spending on high-tech output provides a smaller boost to domestic employment and labor income than spending on property-sector output or the services sector. High-tech expansion can therefore support GDP growth without delivering a proportionate improvement in household income and consumption.

Exhibit 4: Each Yuan Spent on High-tech Industries Supports Less Domestic Value Added, Labour Compensation, and Employment Than Property or Services Sector



Source: NBS, Goldman Sachs Global Investment Research

Sizing High-tech's Growth Contributions

We next assess the contribution of high-tech sectors to real GDP growth and contrast it with that of the property sector.[3]

Following our previous approach, we combine our estimates of domestic value-added ratio with sectoral production data to capture the domestic value added generated by high-tech industries. This includes both these sectors' direct contribution and the value-added created by domestic suppliers along their production chains.

High-tech sectors have made a sizeable contribution to China's real GDP growth (Exhibit

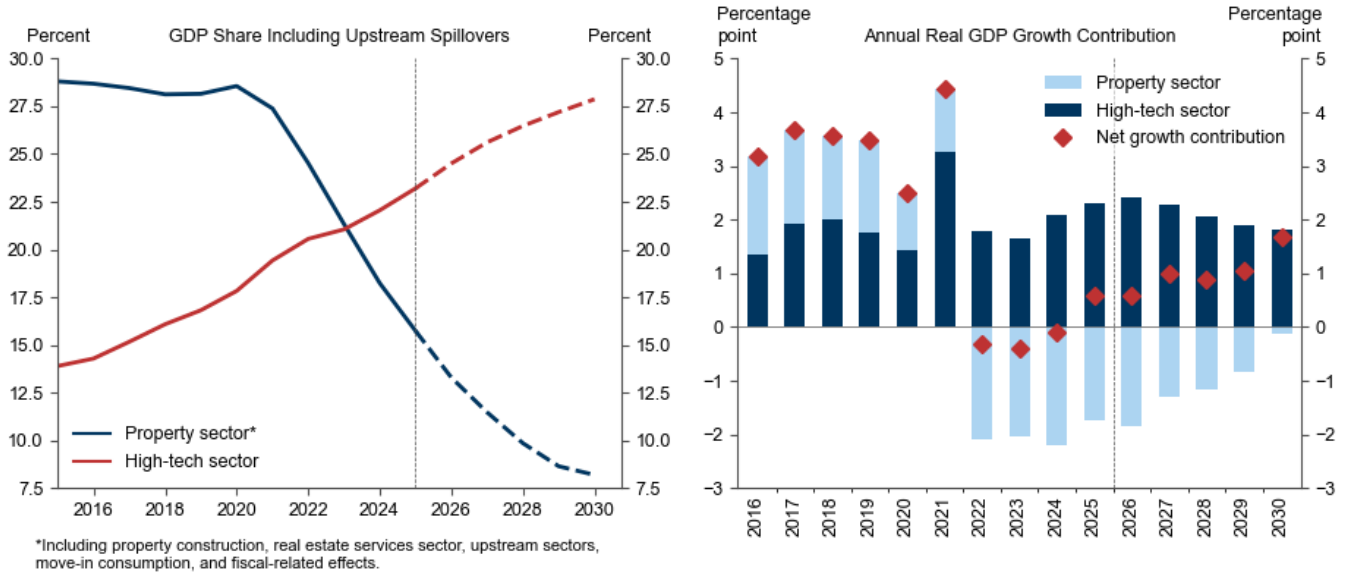
[3] In this analysis, the property sector includes both residential property and non-residential property. Property sector's contribution to GDP includes five channels: construction, real estate services, upstream industries, move-in-related consumption and land sales related fiscal effects.

5). In 2016, high-tech sectors accounted for less than 15% of China's real GDP[4], much less than the 25-30% share of the property sector. We estimate that they contributed an average of 1.9pp to annual real GDP growth over 2016–2025. By 2025, high-tech sectors' GDP share has climbed to 23% whereas the property sector's GDP share has fallen to around 15% after five years of sharp contraction.

Looking ahead, our baseline projects a sustained contribution of around 2pp annually from high-tech sectors over 2026-30. Our key assumption remains that these industries will grow faster than the broader economy, particularly telecom, IT and software services. Our relatively constructive outlook for real exports supports continued growth in high-tech manufacturing sectors, while continued demand for AI compute and digitalization should continue to help IT, telecom and software services sector. Under these assumptions, high-tech sectors' direct share of GDP could rise to around 28% by 2030, while we expect the property sector's share of GDP should stabilize at close to 8% from 2030 onward.

Taken together, while the property downturn is likely to persist in the coming years, its drag on growth should gradually diminish as the sector's share of the economy has fallen meaningfully. The combined contribution of property and high-tech sectors should therefore become more positive over time.

Exhibit 5: High-tech Sectors Sustain a Sizable Growth Contribution As the Property Drag Gradually Diminishes



Source: Haver Analytics, NBS, Goldman Sachs Global Investment Research

Six Networks and Data Center Construction: Quantifying the Investment and Growth Uplift

Data center construction is a key component of China's AI push and an important part of the "Six Networks" investment initiative outlined in the 15th FYP. The initiative spans six areas: water, new-type power grids, computing power, next-generation communications, urban underground pipelines and logistics networks.

The initiative has received renewed policy emphasis, with the July Politburo meeting

[4] Includes the high-tech sector's direct GDP share and the GDP share of its upstream suppliers.

calling for progress on the "Six Networks" and the late-September State Council meeting urging an accelerated start to major projects. In this section, we estimate investment across the "Six Networks" during 15th FYP to assess the increase in spending implied by official policy guidance and its potential implications for FAI growth. In particular, we estimate the growth contribution of data center construction, a key pillar of the computing network, over the next few years.

Overall, we estimate that total investment in the "Six Networks" will increase from RMB24.9tn during the 14th FYP (2020-25) to RMB28.4tn during the 15th FYP (2026-30), a 14% increase, equivalent to 2.7% annualized growth, using the following methodology[5] (Exhibit 6).

- **Power grids**: We aggregate State Grid and China Southern Power Grid investment during 2021–2025 and use their 2026-30 investment plans and recent growth trends to project spending through 2030, cross-checking against the NDRC's projection of more than RMB5tn in new-type grid investment.
- **Water networks**: We use water conservancy construction investment as a proxy, extrapolating the average annual increase in investment over recent years and cross-checking against the official estimate of more than RMB6tn in national water network investment needs.
- **Urban underground pipelines**: We derive an implied unit cost from the NDRC's projected RMB5tn investment and 770,000km of planned construction and renovation, then apply that cost (in nominal terms) to the 840,000km of underground pipelines renovated during the 14th Five-Year Plan to estimate historical investment.
- **Communication networks**: We aggregate historical capex and 2026 budgets of China's three major telecom operators (China Mobile, China Telecom, and China Unicom), and project 2027–2030 spending based on recent investment trends.
- **Logistics networks**: We use FAI in cargo handling and warehousing as a proxy and extrapolate its historical trend.
- **Computing power networks**: We use the estimated RMB4tn in additional direct investment during the 15th FYP, based on an institutional estimate cited by the NDRC, and our equity analysts' forecasts for data center capacity growth to infer investment during the preceding five-year period. We explain these calculations in more detail in the next two paragraphs and Exhibit 7.

[5] Where comparable official figures are unavailable, we construct estimates using historical investment data, corporate capex, sector-level FAI, and project construction metrics.

Exhibit 6: We Estimate an Increase of RMB 3.5tn Investment on "Six Network" Projects From 2021-25 to 2026-30

Estimated total investment (RMB tn)	Example Projects	14th FYP (2021-25)	15th FYP (2026-30)
Logistics Network	Logistics hubs	5.5	6.9
Water Network	Water diversion, reserviors	5.7	6.4
Power Grid	UHV transmission	3.3	5.0
Underground Pipeline	Water, sewage, gas pipelines	5.5	5.0
Computing Power	Data center	3.2	4.0
Communication Network	5G/6G network	1.6	1.1
SUM		**24.9**	**28.4**

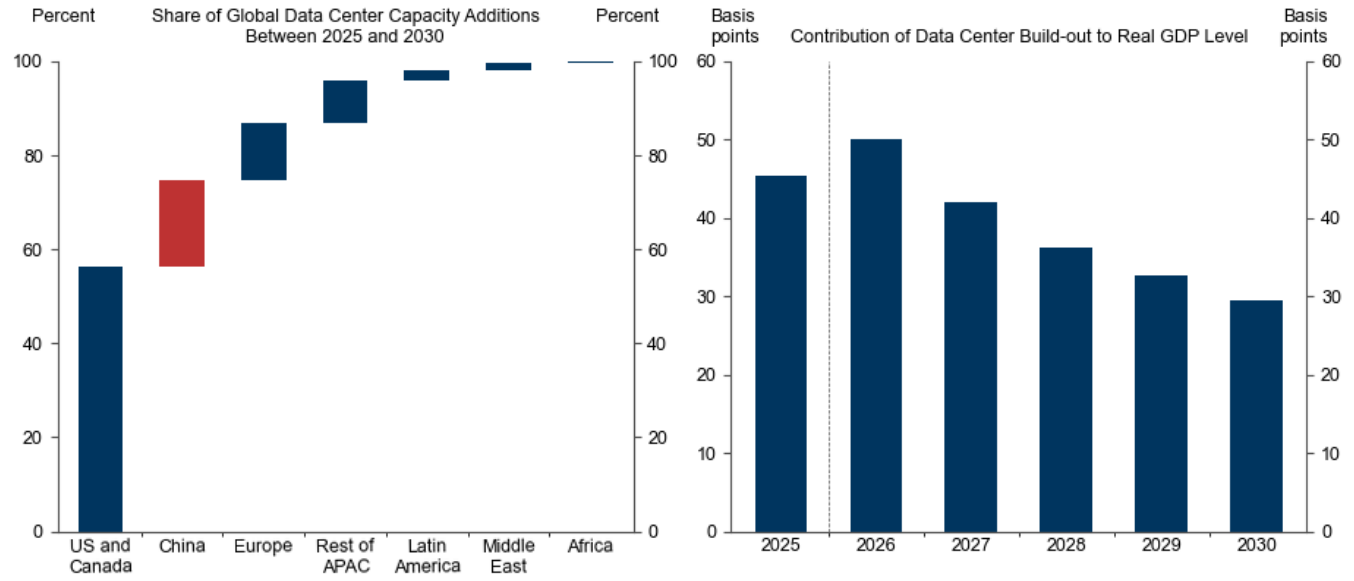
Source: Goldman Sachs Global Investment Research, Government website

A key component of computing network investment is the build-out of data centers. Under the "East Data, West Computing" strategy and the broader coordination of computing resources and power, energy-rich hubs in Western China are becoming increasingly important for capacity expansion. Ulanqab, in particular, is emerging as one of the Asia-Pacific region's largest and fastest-growing AI-computing clusters. Our equity analysts estimate that China will account for nearly 20% of global data-center capacity additions between 2025 and 2030 (Exhibit 7, left).

To estimate the growth impact of this build-out, we use our equity analysts' projections for additions to China's live data-center capacity. Investment is broadly divided between physical infrastructure—including buildings, power-supply systems, and cooling equipment—and computing hardware, notably servers and AI chips. As a growing share of new capacity is designed for AI inference workloads, computing hardware has become a much larger component of total data-center investment, compared with traditional cloud data centers, where physical infrastructure accounted for the bulk of spending. Unlike many other economies, where a meaningful share of electronic components is imported, China's policy focus on technological self-sufficiency likely means that a larger share of chips used in domestic data centers will be sourced locally especially for state-funded data centers[6]. This should limit import leakage and raise the domestic growth multiplier associated with the investment cycle. Overall, our estimates suggest that data center build-out could add around 40bp to the average *level* of China's GDP between 2026 and 2030.

[6] For instance, Bloomberg reported that China requires at least 80% of the infrastructure and AI chips in its nationwide network of AI data centers to be sourced from domestic suppliers.

Exhibit 7: Data Center Build-out Could Add Around 40bp to Annual Real GDP Level Over 2026–30



Source: Goldman Sachs Global Investment Research, MIIT

Taken together, we estimate that RMB28.4tn in cumulative investment across the "Six Networks" could add around 4% to the average *level* of China's GDP between 2026 and 2030, including around 40bp from data center investment. Yet high-tech sectors account for 23% of GDP, including upstream spillovers, underscoring that China's high-tech development extends well beyond data centers or the "Six Networks".

Chelsea Song and Xinquan Chen

The China Economics Team

Andrew Tilton
+852-2978-1802
andrew.tilton@gs.com
Goldman Sachs (Asia) L.L.C.

Hui Shan
+852-2978-6634
hui.shan@gs.com
Goldman Sachs (Asia) L.L.C.

Lisheng Wang
+852-3966-4004
lisheng.wang@gs.com
Goldman Sachs (Asia) L.L.C.

Xinquan Chen
+852-2978-2418
xinquan.chen@gs.com
Goldman Sachs (Asia) L.L.C.

Yuting Yang
+852-2978-7283
yuting.y.yang@gs.com
Goldman Sachs (Asia) L.L.C.

Chelsea Song
+852-2978-0106
chelsea.song@gs.com
Goldman Sachs (Asia) L.L.C.

Disclosure Appendix

Reg AC

We, Chelsea Song and Xinquan Chen, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Chelsea Song Goldman Sachs (Asia) L.L.C., Xinquan Chen Goldman Sachs (Asia) L.L.C..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number 60634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are

available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or asil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou asil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für tungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g.,

marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. You may not use, extract, download or retrieve this information, in whole or in part, to train, finetune, adjust, influence the weights or behavior of, or otherwise develop or improve a machine learning or artificial intelligence model or system, or provide or reproduce this information, in whole or in part, as a prompt or input to any such system without the express written consent of Goldman Sachs. Furthermore, you may not bulk download this information, or employ any robot, spider, crawler, scraper, automated agent, or other electronic means to scrape, harvest, extract, or retrieve this information, without the express written consent of Goldman Sachs. A breach of the above terms may result in termination of your right to access this service.